CHL5402 Epidemiologic Methods – Tutorial example for confounding and effect modification in published results

This is a cue for discussion and students are asked to NOT look up the original whole study. What the authors say and what we would conclude may be very different! The exercise requires that you use only the available information, below. Students are asked to apply the concepts for detection of confounding and effect modification from the epidemiology courses and resources within the MPH. Students are permitted to look up definitions of terms within the abstract that they don't understand, of course, and should refer (as required) to course notes, lecture slides and required readings from Epi I/Epi II and Biostats I and II.

Spousal Loss and Cognitive Function in Later Life: A 25-year Follow-up in the AGES-Reykjavik Study. Vidarsdottir H, Fang F, Chang M, Aspelund T, Fall K, Jonsdottir MK, Jonsson PV, Cotch MF, Harris TB, Launer LJ, Gudnason V, Valdimarsdottir U. Am J Epidemiol. 2014 Mar 15;179(6):674-83. doi: 10.1093/aje/kwt321. Epub 2014 Jan 19.

Abstract: The aim of this study was to investigate the associations between loss of a life partner and the development of dementia and decline in cognitive function in later life. We used an Icelandic cohort of 4,370 participants in the Age, Gene/Environment Susceptibility-Reykjavik Study who were living as married in 1978 (born in 1907-1935) and were either still married (unexposed cohort) or widowed (exposed cohort) at follow-up (in 2002-2006). We ascertained history of marital status and spouse's death by record linkage to the Registry of the Total Population, Statistics Iceland. The outcome measures were as follows: 1) dementia and mild cognitive impairment; and 2) memory, speed of processing, and executive function. During the observation period, 3,007 individuals remained married and 1,363 lost a spouse through death. [rest of abstract withheld, in context].

Table 2. Multivariable Logistic Regression–Derived Odds Ratios for Associations Between Spousal Loss and Mild Cognitive Impairment in the AGES-Reykjavik Study, 2002–2006

Subgroup	Mild Cognitive Impairment (n = 404 cases)							
	No.	%	Crude		Age Adjusted		Full Model[a]	
			OR	95% CI	OR	95% CI	OR	95% CI
Total	3,909	100						
Spouse alive	2,758	69	1.00	Referent	1.00	Referent	1.00	Referent
Spouse deceased	1,151	31	1.50	1.21, 1.85	0.93	0.74, 1.18	0.99	0.76, 1.30
Men								
Spouse alive	1,494	84	1.00	Referent	1.00	Referent	1.00	Referent
Spouse deceased	280	16	1.65	1.17, 2.32	1.08	0.75, 1.55	0.84	0.55, 1.27
Women								
Spouse alive	1,264	59	1.00	Referent	1.00	Referent	1.00	Referent
Spouse deceased	871	41	2.06	1.50, 2.81	1.16	0.82, 1.64	1.11	0.77, 1.60

[a] **Full Model Adjusted** for age, sex, ApoE E4 (genetic) status, education, smoking (ever or never), midlife hypertension, midlife regular physical activity (yes or no), and midlife body mass index (weight (kg)/height (m)2).
ApoE E4 carriers have a higher risk of developing dementia. Because the ApoE E2 allele may play a protective role in dementia,

[Downloaded and adapted from http://aje.oxfordjournals.org/].